Open Logic N1i normalization: authored core and context

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This derivative reader collects the author's segment definitions, permutation conversions, reduction conversions, and normalization proof from the fixed source. The selected target is N1i normalization. Author exercises are retained without solutions; the subsequent N2i statement and the strong-normalization remark remain author context and are not additional selected questions. The mathematical text and proof trees are unchanged. Source licence: Creative Commons Attribution 4.0 International.

Segments and Cuts

An introduction rule followed by an elimination rule is a detour in a derivation, and to normalize a derivation is to eliminate such detours. However, the ∨Elim and ∃Elim make even the definition of "detour we want to eliminate" a bit complicated. The problem is this. Because of these rules, it's not guaranteed that in a detour the elimination rule *immediately* follows the introduction rule. There could be an ∨Elim or ∃Elim in between, e.g.,

$$\dfrac{\dfrac{\exists x\,\chi(x) \qquad \overset{x}{\dfrac{\overset{[\varphi]^x}{\vdots}\ \psi}{\varphi \to \psi}}\ {\to}\text{Intro}}{\varphi \to \psi}\ \exists\text{Elim}^{\,y} \qquad \vdots\ \varphi}{\psi}\ {\to}\text{Elim}$$

In fact, any number of ∨Elim or ∃Elim can separate the →Intro from the →Elim, e.g.,

$$\dfrac{\dfrac{\exists x\,\chi(x) \qquad \dfrac{\theta \vee \alpha \qquad \overset{x}{\dfrac{\overset{[\varphi]^x}{\vdots}\ \psi}{\varphi \to \psi}}\ {\to}\text{Intro} \qquad \overset{x}{\dfrac{\overset{[\varphi]^x}{\vdots}\ \psi}{\varphi \to \psi}}\ {\to}\text{Intro}}{\varphi \to \psi}\ \vee\text{Elim}}{\varphi \to \psi}\ \exists\text{Elim}^{\,y} \qquad \vdots\ \varphi}{\psi}\ {\to}\text{Elim}$$

This example shows that the same →Elim might be involved in multiple detours, even, as it follows more than one →Intro. To account for all these possibilities, we define detours using certain kinds of sequences of formula occurrences in an **N1i**-proof called *segments*. In our example, it's a sequence of occurrences of $\varphi \to \psi$, where a minor premise of ∨Elim or ∃Elim follows its conclusion. In the example, there are two such segments, each containing three occurrences of $\varphi \to \psi$. Here is the official definition.

Definition 0.1. A *segment* in a **N1i** derivation δ is a sequence of occurrences φ_1, ..., φ_n of the same formula φ in δ where

1. φ_1 is not the conclusion of ∨Elim or ∃Elim;
2. φ_n is not a minor premise of ∨Elim or ∃Elim;
3. For $i = 1, \ldots, n-1$, φ_i is a minor premise of ∨Elim or ∃Elim and A_{i+1} is the conclusion of that inference.

The *length* of that segment is n.

Not every segment is a target for elimination (a detour). The ones that are are called *cuts*.

Definition 0.2. A *cut* is a segment in which φ_n is the major premise of an elimination inference, and either $n = 1$ and $\varphi_1 = \varphi_n$ is the conclusion of an introduction inference or of $\bot_I$, or $n > 1$.

Note that a cut segment need not begin with the conclusion of an introduction rule. All that's required is that it ends with the major premise of an elimination rule. However, a segment of length 1 does have to be the conclusions of introduction rules to count as a cut.

Definition 0.3. The *rank* of a cut is $\mathrm{dp}(\varphi)$. The *cut rank* $\mathrm{cr}(\delta)$ of a derivation δ is the maximal rank of cuts in δ. A cut is *maximal* in δ if its rank is $\mathrm{cr}(\delta)$, i.e., it has maximal rank. The *cut length* of δ is the sum of the lengths of all its maximal cuts.

Permutation Conversions

Cuts in **N1i** are segments that end in the major premise of an elimination rule. One case in the normalization proof is to reduce the length of maximal cuts, and with it the cut length of the derivation. A cut of length > 1 can end in many ways depending on whether the last formula occurrence of φ in the cut is the conclusion of ($\vee$Elim or $\exists$Elim) and depending on the elimination inference it is the major premise of.

For instance, if the inferences involved are $\vee$Elim and $\wedge$Elim, $\varphi \equiv \psi \vee \chi$ and there is a sub-derivation δ_1 that ends in:

$$\dfrac{\dfrac{\theta \vee \alpha \quad \begin{matrix}[\theta]^x \\ \vdots\,\delta_3 \\ \psi \wedge \chi\end{matrix} \quad \begin{matrix}[\alpha]^y \\ \vdots\,\delta_4 \\ \psi \wedge \chi\end{matrix}}{\psi \wedge \chi}\,{\scriptstyle x\,y}\ \vee\mathrm{Elim}}{\psi}\ \wedge\mathrm{Elim}_1 \qquad \vdots\,\delta_2$$

The last formula occurrence φ_n in the cut is the conclusion of $\vee$Elim, the lower $\psi \wedge \chi$. The second-to-last formula occurrence φ_{n-1} is one of the minor premises of $\vee$Elim.

To reduce the length of this cut, we can switch the order ("permute") of the $\vee$Elim and $\wedge$Elim inferences. That is, we replace the subproof δ_1 in δ by the following:

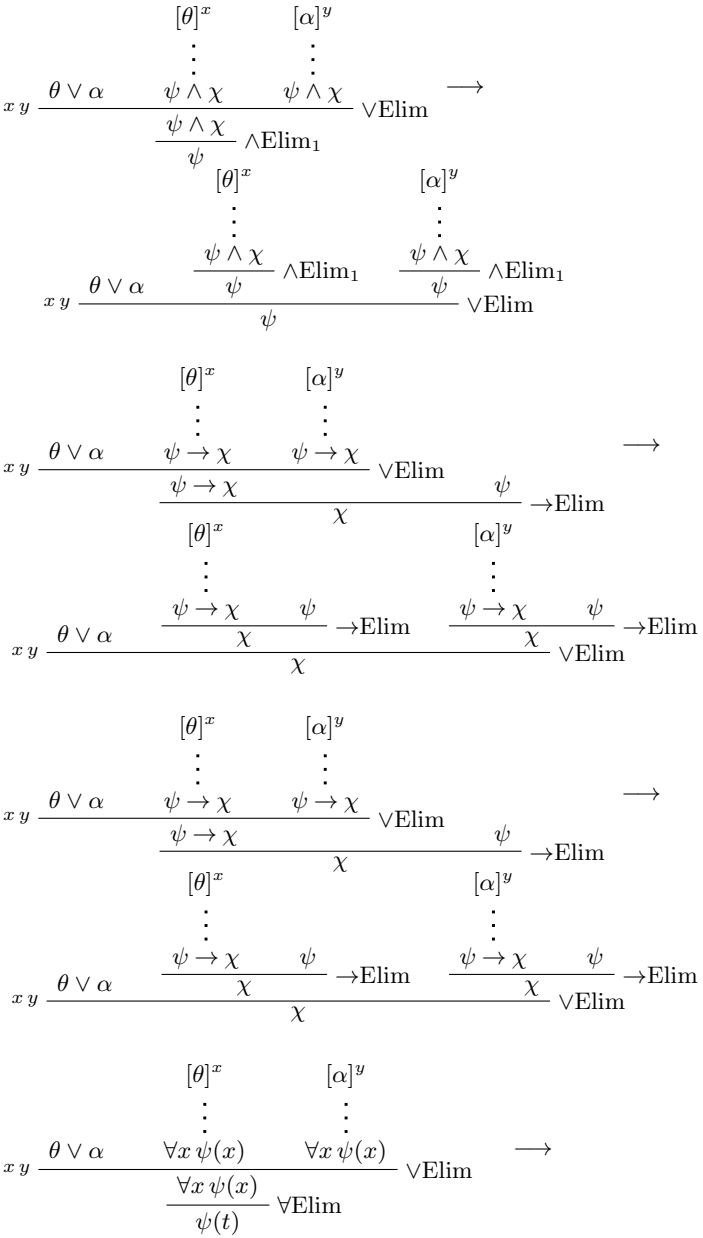
$$\dfrac{\theta \vee \alpha \quad \begin{matrix}[\theta]^x \\ \vdots\,\delta_3 \\ \psi \wedge \chi\end{matrix} \quad \begin{matrix}[\alpha]^y \\ \vdots\,\delta_4 \\ \psi \wedge \chi\end{matrix}}{\psi}\,{\scriptstyle x\,y}\ \vee\mathrm{Elim} \qquad \dfrac{\vdots\,\delta_2}{\psi \wedge \chi}\ \wedge\mathrm{Elim}_1 \qquad \dfrac{\vdots\,\delta_2}{\psi \wedge \chi}\ \wedge\mathrm{Elim}_1$$

The cuts that ended in the premise of $\wedge$Elim below $\vee$Elim now end in the premise of one of the $\wedge$Elim inferences above the minor premises of $\vee$Elim, i.e., the length of each has decreased by 1.

Moreover, any cut that lies entirely within δ_2, δ_3, or δ_4, or entirely outside δ_1 in δ is unchanged: it still involves the same formulas, goes through the same inferences, so in particular has the same rank and length as the corresponding cut in δ. So the cut rank of the new derivation has not increased, and the cut length of the new derivation is less than that of the old derivation.

Problem 0.1. After a permutation conversion, the length of at least one cut is reduced by 1. If we permute $\wedge$Elim across an $\vee$Elim, there are actually two cut segments the length of which decrease, so in this case the cut length of the derivation decreases by at least 2. Can the cut length of a derivation decrease by more than 2 by a single such permutation? Can the cut *rank* decrease?

The process of moving an elimination rule above an $\vee$Elim or $\exists$Elim rule is called a *permutation conversion*. The patterns for some of the rule pairs is given in 1. (The remaining are left as exercises.)



$$\dfrac{\theta \vee \alpha \qquad \dfrac{\begin{array}{c}[\theta]^x\\ \vdots\\ \forall x\,\psi(x)\end{array}}{\psi(t)}\,\forall\text{Elim} \qquad \dfrac{\begin{array}{c}[\alpha]^y\\ \vdots\\ \forall x\,\psi(x)\end{array}}{\psi(t)}\,\forall\text{Elim}}{\psi(t)}\,{\scriptstyle x\,y}\vee\text{Elim}$$

Table 1: Some permutation conversions for **N1i**.

Problem 0.2. Give the permutation conversions for ∨Elim followed by ∨Elim or ¬Elim.

Problem 0.3. Give the permutation conversions for ∃Elim followed by ∃Elim. Explain why no eigenvariable condition is violated in the latter.

We should convince ourselves that when applying a permutation conversion, we don't perhaps violate an eigenvariable condition. Consider

$$\dfrac{\dfrac{\theta \vee \alpha \qquad \begin{array}{c}[\theta]^x\\ \vdots\\ \exists x\,\psi(x)\end{array}\,\delta_3 \qquad \begin{array}{c}[\alpha]^y\\ \vdots\\ \exists x\,\psi(x)\end{array}\,\delta_4}{\exists x\,\psi(x)}\,{\scriptstyle x\,y}\vee\text{Elim} \qquad \begin{array}{c}[\psi(c)]^z\\ \vdots\\ \chi\end{array}\,\delta_5}{\chi}\,{\scriptstyle z}\,\exists\text{Elim}\quad \delta_2$$

If we replace this by

$$\dfrac{\theta \vee \alpha \qquad \dfrac{\begin{array}{c}[\theta]^x\\ \vdots\\ \exists x\,\psi(x)\end{array}\,\delta_3 \qquad \begin{array}{c}[\psi(c)]^z\\ \vdots\\ \chi\end{array}\,\delta_5}{\chi}\,{\scriptstyle z}\,\exists\text{Elim} \qquad \dfrac{\begin{array}{c}[\alpha]^y\\ \vdots\\ \exists x\,\psi(x)\end{array}\,\delta_4 \qquad \begin{array}{c}[\psi(c)]^z\\ \vdots\\ \chi\end{array}\,\delta_5}{\chi}\,{\scriptstyle z}\,\exists\text{Elim}}{\chi}\,{\scriptstyle x\,y}\vee\text{Elim}\quad \delta_2$$

we might worry that the eigenvariable c might occur in, say, θ. After all, the assumption θ is discharged above the original ∃Elim and so in the original derivation it does not occur in an assumption that's open above the major premise of the ∃Elim inference, and so there the eigenvariable condition is satisfied. But in the new derivation, θ is not discharged until after the ∃Elim inferences, and so the eigenvariable condition would be violated. Recall, however, that we assume our derivations are regular: every eigenvariable is the eigenvariable of a single inference and only occurs above the minor premise of ∃Elim in the derivation. In particular, c cannot occur in δ_3 or δ_4.

This example shows something else: Note that the new derivation has two copies of δ_5. This means that if δ_5 contains a maximal cut, we have *not* reduced the cut length!Thus we must take care that we apply a permutation conversion only when we know that there is no maximal cut in δ_5. We will ensure this by always selecting a *topmost* maximal cut, that is a maximal cut such that the sub-derivation (in this case, δ_1) contains no maximal cut that ends above the last inference in the sub-derivation. This rules out the possibility that there are other maximal cuts in δ_5 (or indeed in δ_2, δ_3, or δ_4).

Reduction Conversions

If a cut in a derivation in **N1i** has length 1, it is a segment consisting of a single formula occurrence, and that formula is both the conclusion of an introduction inference and the major premise of an elimination inference. In the normalization proof, such cuts are "reduced" by replacing the sub-derivation ending in such an introduction/elimination

pair by a new derivation in which the pair of inferences has been eliminated. Such a replacement may result in new cuts. However, we can convince ourselves that the new cuts have lower rank than the cut being reduced. Thus, the new derivation either has lower cut length (because a maximal cut of length 1 has been removed) or the cut rank of the derivation has been reduced (if the maximal cut in question was the only maximal cut).

For instance, if the inferences involved are →Intro and →Elim, $\varphi \equiv \psi \to \chi$ and there is a sub-derivation δ_1 that ends in:

$$\dfrac{\dfrac{\begin{array}{c}[\psi]^x\\ \vdots\,\delta_2\\ \chi\end{array}}{\psi \to \chi}\,{\scriptstyle x}\,{\to}\text{Intro} \qquad \begin{array}{c}\vdots\,\delta_3\\ \psi\end{array}}{\chi}\,{\to}\text{Elim}$$

To remove this cut we replace the subproof δ_1 in δ by the following:

$$\begin{array}{c}\vdots\,\delta_3\\ \psi\\ \vdots\,\delta_2[\delta_3/\psi^x]\\ \chi\end{array}$$

By $\delta_2[\delta_3/\psi^x]$ we mean the derivation that results from δ_2 by replacing every assumption ψ that's labelled x with the entire derivation δ_3. This derivation no longer has open assumptions of the form ψ with label x. It has no other assumptions labelled x, since we assume, as usual, that no two formulas occur as assumptions with the same label. Hence, the new derivation has the same open assumptions as δ_1, every inference in δ_2 that discharges assumptions discharges the same assumptions in $\delta_2[\delta_3/\psi^x]$, and every inference in δ_3 that discharges assumptions results in possibly multiple copies, which discharge the corresponding assumptions in $\delta_2[\delta_3/\psi^x]$.

Any cut that lies entirely within δ_2, δ_3, or δ_4, or entirely outside δ_1 in δ is unchanged: it still involves the same formulas, goes through the same inferences, so in particular has the same rank and length as the corresponding cut in δ. We have removed a maximal cut of length 1 and so either have reduced the cut length, or reduced the cut rank (if the cut length of δ was 1 and this was the only maximal cut).

However, two things may happen:

1. By replacing all assumptions ψ^x by δ_3, we have multiplied all the cuts in δ_3. If some of those were maximal cuts, the cut length of the new derivation may be greater than the cut length of the old derivation. We have to guarantee that this does not happen. We do so by applying reduction conversions only to *rightmost* maximal cuts, that is, to cuts which are such that no maximal cut lies on a branch through δ to the right of the cut being reduced. If there were a cut in δ_3 it would be to the right of the cut being reduced here, and so we wouldn't be working on a rightmost cut. By always selecting rightmost cuts, we guarantee that either we decrease the cut length of the derivation, or even reduce the cut rank.

2. If an assumption ψ^x in δ_2 is the major premise of an elimination inference, and the last inference of δ_3 is ∨Elim, ∃Elim, or an introduction inference, we have generated a new cut by grafting δ_3 onto δ_2 at such an assumption. What was just an assumption in δ_2 is now a cut of length 1 (the conclusion of an introduction inference and major premise of an elimination inference) or the last formula occurrence of a cut segment (major premise of an elimination inference and conclusion of ∨Elim or ∃Elim). If B^x was a minor premise of ∨Elim or ∃Elim, it already was the first formula occurrence of a segment, and possibly of a cut segment. In the new derivation that segment may now be longer. However, observe that the formula making up this new cut or existing cut segment is ψ, and $\mathrm{dp}(\psi) < \mathrm{dp}(\psi \to \chi)$. Hence, although we have potentially added cuts or increased the length of cuts, none of these are maximal cuts. Thus we have either decreased the cut rank, or, if there are cuts of the same rank remaining, have decreased the cut length.

The process of reducing cut of length 1 is called a *reduction conversion* (or *detour conversion*). The patterns for some of the rule pairs is given in 2. (The remaining are left as exercises.)

Problem 0.4. Give the reduction conversion for ¬Intro followed by ¬Elim.

There is one case still to be considered. The definition of cut includes the case where the length of the cut is 1 and φ is the conclusion of $\bot_I$ as well as the major premise of an elimination inference. This case is dealt with in a similar way as when φ is the conclusion of an introduction rule. For instance, replace

$$\dfrac{\dfrac{\begin{array}{c}\vdots\,\delta_2\\ \bot\end{array}}{\psi\to\chi}\quad\begin{array}{c}\vdots\,\delta_3\\ \psi\end{array}}{\chi}\to\text{Elim}\quad\text{by}\quad\begin{array}{c}\vdots\,\delta_2\\ \bot\\ \hline \chi\end{array}$$

We must be a little careful in the cases where $\varphi \equiv (\psi \vee \chi)$ or $\varphi \equiv \exists x\, \psi(x)$. E.g., if we replaced

$$\dfrac{\dfrac{\begin{array}{c}\vdots\,\delta_2\\ \bot\end{array}}{\psi\vee\chi}\,\bot_I\quad\begin{array}{c}[\psi]^x\\ \vdots\,\delta_3\\ \theta\end{array}\quad\begin{array}{c}[\chi]^x\\ \vdots\,\delta_4\\ \theta\end{array}}{\theta}\vee\text{Elim}\quad\text{by}\quad\dfrac{\begin{array}{c}\vdots\,\delta_2\\ \bot\end{array}}{\theta}\,\bot_I$$

we might be introducing a new cut with cut formula θ, and there is no guarantee that $\mathrm{dp}(\theta) < \mathrm{dp}(\psi \vee \chi)$! So we must proceed the same way here as we did for an ∨Intro/∨Elim reduction and replace with

$$\begin{array}{c}\vdots\,\delta_2\\ \dfrac{\bot}{\psi}\,\bot_I\\ \vdots\,\delta_3\\ \theta\end{array}\qquad\text{or}\qquad\begin{array}{c}\vdots\,\delta_2\\ \dfrac{\bot}{\chi}\,\bot_I\\ \vdots\,\delta_4\\ \theta\end{array}$$

The Normalization Theorem

A derivation is in *normal* form if it has no cut segments. Clearly this is the case iff we can apply neither a permutation conversion nor a reduction conversion to it. To *normalize* a derivation means to transform it into a normal proof by applying permutation and reduction conversions until no cuts remain. That this is always possible is the *normalization theorem*.

Theorem 0.4. *Any* **N1i**-*derivation δ normalizes, i.e., can be transformed into a derivation δ' without cuts.*

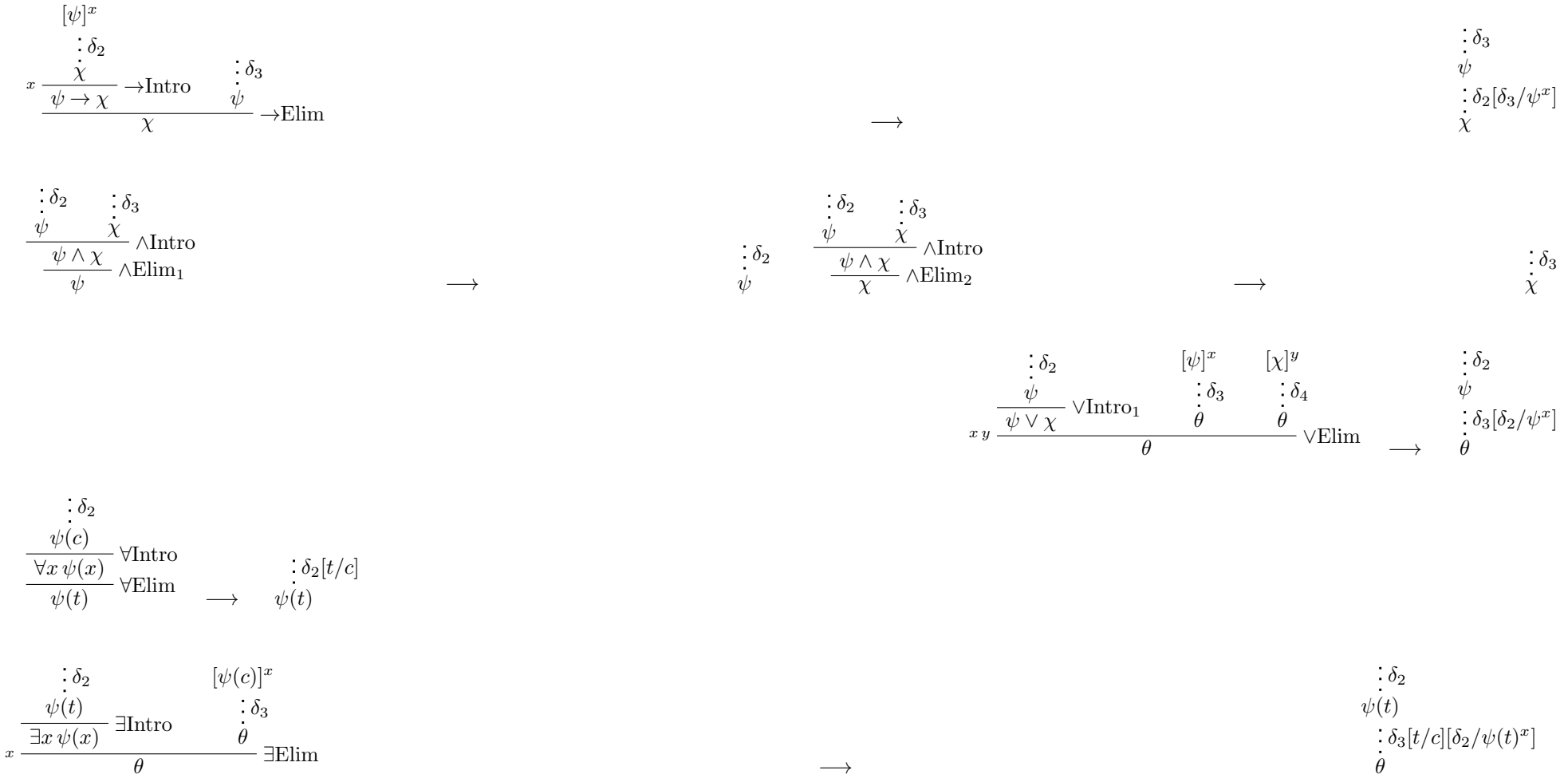


Table 2: Some reduction conversions for **N1i**.

Proof. By induction on the cut rank and length of δ. The induction hypothesis is that any derivation which either has lower cut rank, or the same cut rank but lower cut length, normalizes.

If the cut length is 0, there are no cuts, δ is already normal. So assume that cut rank and cut length of δ are > 0. Pick a topmost, rightmost maximal cut segment. If the cut segment has length > 1, apply a permutation conversion. If the cut segment has length 1, apply the corresponding reduction conversion. This results in a new derivation which either has the same cut rank but a lower cut length, or a lower cut rank, and the inductive hypothesis applies. □

The strategy used in the above proof (always deal with a topmost rightmost maximal cut) suggests a specific order in which conversions must be applied to a derivation in order to result in a normal derivation. Surprisingly enough, the order actually does not matter. Applying permutation and reduction conversions in any order eventually results in a normal proof.

The definition of segment and cut can easily be formulated for **N2i**, and the permutation and reduction conversions for **N1i** directly translate to **N2i**. Consequently, the normalization theorem also holds for **N2i**.

Theorem 0.5. *Any* **N2i***-derivation δ normalizes, i.e., can be transformed into a derivation δ' without cuts.*